25901190 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCSC201 - COMPUTER NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define syntax and Semantics in a network protocol [K] / [CO1]
2. List any two characteristics of circuit switching. [K] / [CO2]
3. Identify one technique used for error correction. [U] / [CO3]
4. State the range of port numbers available in UDP. [K] / [CO4]
5. Define LCP in PPP. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Differentiate between LAN and WAN. [K] / [CO1]

7. Explain the working of a connection-oriented service. [K] / [CO2]

8. Describe the features of IPv4 protocol. [U] / [CO3]

9. Explain the working of TELNET in a computer network. [U] / [CO4]

10. Outline the layered architecture of the OSI model. [K] / [CO1]

11. Explain how a datagram network delivers packets without a dedicated path. [U] / [CO2]

12. Show how Go-Back-N ARQ handles frame errors. [A] / [CO3]

13. Explain the role of protocol in a URL. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. State the different types of network topologies with its relevant features. [K] / [CO1]

15. Explain the classification of routing algorithms. [U] / [CO3]

16. Explain SMTP in detail. [U] / [CO4]

[15x2 = 30]